Surya Ravi

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About Me

Engineer with cross-domain knowledge spanning cybersecurity, computer science, AI and system design, backed by degrees in mechatronics engineering and computer animation. Experienced in threat exposure management, SOAR automation, vulnerability management, bug bounty hunting and AI red teaming, and in building security tooling that pairs static analysis and LLM agents with verified, exploit-backed findings.

Experience

Orbia Dec. 2025 – Present
Senior Analyst – Threat Exposure Management *Pune, Maharashtra*

- Designed, developed and currently maintain **18** end-to-end automations using **Tines SOAR** covering threat monitoring, alert triage, vulnerability management workflows and false positive reclassification, improving response efficiency, decreasing manual work hours and reducing the number of false positive reports.
- Validated comprehensive penetration testing reports, triaging bugs and ensuring accurate documentation of vulnerabilities, proof-of-concepts, and remediation recommendations in **Intigriti** and **HackerOne**.
- Developed **5 CrowdStrike Falcon Fusion SOAR** workflows to automate security response and remediation actions, enabling faster investigations, decision-making and endpoint response while reducing manual analyst intervention.
- Executed **2** white-box penetration tests on internal web applications, identifying security gaps and collaborating with engineering teams for remediation.
- Developed and optimized security orchestration workflows integrating **CrowdStrike**, **Tenable**, **ServiceNow**, **Assetnote** and **Proofpoint** to prioritize vulnerabilities in endpoints and application environments.
- Built API-driven security automation workflows using **REST APIs**, **Postman**, **JSON** and **Python**, enabling integration and processing of security data across disparate security platforms.
- Managed the end-to-end vulnerability lifecycle, from identification and risk assessment through remediation tracking, validation/retesting, stakeholder coordination, and compliance reporting.
- Performed remediation validation and retesting to verify successful patch implementation and confirm vulnerability resolution in **8** internal applications.

Alice 2026
AI Red Team Researcher – Freelance *Remote*

- Conducted **AI red-team and adversarial security testing using Alice**, developing attack scenarios to evaluate model behavior, prompt-injection resistance, guardrails, and security boundaries.
- Built and iterated on **automated security-testing harnesses** for adversarial prompt execution, response capture, reproducibility, and regression testing across attack scenarios.
- Identified and validated **guardrail bypasses and unintended model behaviors**, documenting reproducible attack paths, security impact, and defensive improvements.
- Applied **prompt injection**, **instruction-hierarchy manipulation**, model steering and output-format coercion techniques to probe LLM applications for jailbreaks, system-prompt leakage and sensitive data exposure.
- Mapped findings to the **OWASP Top 10 for LLM Applications** and reported them with severity ratings, reproduction steps and mitigation guidance for model and product teams.

HackerOne 2025
Bug Bounty Hunter – Freelance *Remote*

- Hunted for web and API vulnerabilities in public **HackerOne** programs, focusing on access control, authentication, rate limiting and exposed staging infrastructure.
- Reported **3** High vulnerabilities and **4** Medium vulnerabilities with auth, API and injection vulnerabilities, documenting reproduction steps, explaining security impact, and summarizing remediation recommendations.
- Used **Burp Suite** for DAST and developed automated pipelines for SAST reconnaissance, request tampering and validation of findings, staying within program scope and disclosure rules.
- Performed **CVE variant analysis**, studying the root causes of published CVEs across different vulnerability classes and hunting for variants of the same flaw patterns in in-scope targets.

Hackup Technology Aug. 2025 – Nov. 2025
Cybersecurity Analyst Intern *Coimbatore, Tamil Nadu*

- Conducted hands-on web application security testing using **Burp Suite**, analyzing application behavior, identifying security weaknesses, and developing practical understanding of vulnerability discovery and validation.
- Performed network traffic analysis using **Wireshark**, examining protocols, packet flows, and network communications to strengthen understanding of security monitoring and detection of anomalous activity.
- Gained practical exposure to SOC operations, threat monitoring, vulnerability management, and security assessments, developing foundational experience in identifying, analyzing, and responding to security issues.
- Delivered technical seminars across **3** colleges covering **Linux** fundamentals, command-line operations, and cybersecurity concepts, translating technical security topics into practical guidance for students.
- Developed practical knowledge of web application vulnerabilities aligned with **PortSwigger Web Security Academy**, including SQL Injection, IDOR, API security flaws, NoSQL Injection, Reflected and Stored Cross-Site Scripting, Authentication and Access Control issues, and other OWASP-aligned vulnerabilities, with hands-on focus on exploitation, validation, and remediation techniques.

Education

Kingston University — Sep. 2023 – Sep. 2024
Master's Degree in Computer Animation (School of Computer Science & Mathematics) — *London, United Kingdom*
Award: Distinction

Anna University — Aug. 2019 – May 2023
Bachelor's in Mechatronics Engineering — *Chennai, Tamil Nadu*
GPA: 8.8

PVV Higher Secondary School — Apr. 2017 – Apr. 2019
HSC: 91.4% & SSLC: 94.8% — *Coimbatore, Tamil Nadu*

Projects

arc-ai – PoC-generating security agent
- LangGraph + Temporal agent that finds vulnerabilities from source, proves each with a working exploit in an egress-locked sandbox and reports only verified findings; proves 3/3 planted vulnerabilities (SQLi, SSRF, IDOR) and discards unprovable hypotheses, for zero unverified alerts.

taintline – Static taint analyzer
- Go + tree-sitter analyzer tracing user input to SQL injection, command injection and path-traversal sinks in Python and JavaScript, with SARIF output for GitHub Code Scanning; detects 9/9 seeded flaws with 0 false positives on safe twins, backed by 42 engine edge-case tests.

PPx – Prompt Pattern Index
- LangGraph pipeline that classifies prompts across 4 prompt-injection pattern families (prompt extraction, instruction override, data exposure, malicious patterns) and produces a risk score, severity rating and security report.

Attack Surface Mapper
- Application-understanding engine that maps a codebase in 4 discovery stages (framework, routes, authentication, external services) before vulnerability hunting; surfaced 37 routes, 4 admin routes and 2 external services on a sample Express app.

Writeups

- *Alice LLM Security* – LLM-focused security research through prompt injection, instruction-hierarchy manipulation, adversarial prompting, model steering, and output-format coercion while solving an AI security CTF. Awarded Best Technical Write-up. Read
- *QKD* – Research on Quantum Key Distribution and its cryptographic foundations, exploring quantum principles, secure key exchange, eavesdropping detection, and the role of QKD in post-quantum cybersecurity. Read

Skills

Primary Skills: SOAR, Incident Response, MITRE ATT&CK, Threat Intelligence, API Security, OWASP Top 10, Cryptography, Vulnerability Management, Python, Java, HTML, Shell, C
Secondary Skills: Bash, Linux, Penetration Testing, Quantum Cryptography, Tree-sitter, CI/CD, LLM Security, Adversarial Testing, Relational and Non-relational database
Stack: Tines, Burp Suite, LangGraph, Postman, Git, GitHub, VS Code and Docker
Certifications: GCPC, Tines Certified Professional, Postman API Developer Associate, CrowdStrike Falcon Fusion SOAR and LangGraph Foundations
Soft Skills: Problem-Solving, Fast-paced Learning and Cross-functional collaboration
Hobbies: Guitar, Surfing, solving Rubik's Cube & Sudoku, Chess and Reading

Honours & Awards

Name	*Placing*	*Type*	*Scope*	*Awarder*	*Year*
Alice in Wonderland LLM CTF	Top 5	AI Security CTF	Global	Alice	2026
Kriya CTF (Jeopardy)	1st Place	Security CTF	IN	PSG Institute of Technology	2020
IISF Hackathon	1st Place	Hackathon	Global	Ministry of Sci. & Tech.	2020
Robo Sumo	1st Place	Robotics	IN	PSG Institute of Technology	2021
Outstanding Student Award	Academic & Inter-College Event Excellence	Academic Award	IN	Anna University	2023
Certificate of Recognition	Academic Merit	Academic Award	IN	Anna University	2023
Mahatma Gandhi Merit Scholarship	Top 10% of Institution	Scholarship	IN	Anna University	2019, 2020